GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Process System Description**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-004
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Originator	Akinwale Biobaku
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Process System Description

R02	27.09.22	Re-issued for Review	B. Akinwale	M. Ekpunobi	A. Duru	
R01	31.05.22	Issued for Review	B. Akinwale	K. Yusuff	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	31.05.22	Issued for Client Review
R02	27.09.22	Re-issued for Review after incorporating/clarifying TCE comments

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-004

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
AASHTO	American Association of State Highway Transportation Officials
AGEF	Above Ground End Facilities
ALARP	As Low as Reasonably Practicable
API	American Petroleum Institute
ASTM	American Society of Testing and Materials
BOD	Basis Of Design
CP	Cathodic Protection
CPT	Cone Penetrometer Test
CTMS	Custody Transfer Metering Station
DB&B	Double Block and Bleed
DCQ	Daily Contractual Quantity
DEP	Design and Engineering Practice
ESD	Emergency Shutdown
ESDV	Emergency Shutdown Valve
FEED	Front End Engineering Design
FGS	Fire, Gas and Smoke detection systems
GPP	Gas Processing Plant
GTG	Gas Turbine Generators
HAZID	Hazard Identification
HAZOP	Hazard Operability
HEMP	Hazards and Effects Management Process
HFE	Human Factors Engineering
HPU	Hydraulic Power Unit
HRA	Health Risk Assessment
HSES	Health, Safety, Environment and Security
HSSE & SP	Health, Safety, Security, Environment and Social Performance
IEC	International Electro-technical Commission

ISPM	Industry Standard Process Modules
JV	Joint-Venture
LCD	Liquid Crystal Display
LOI	Local Operator Interface
LRL	Lower Range Limit
LRV	Lower Range Value
LIRA	Logistics and Infrastructure Resource Assessment
MAC	Main Automation Contractor
NAG	Non-Associated Gas
OPC	OLE for Process Control
OPPS	Overpressure Protection System
PCD	Process Control Domain
PDMS	Plant Design and Management System
PEPF	Package Engineering, Procurement, and Fabrication
PIG	Pipeline Inspection Gauge
PVC	Polyvinyl Chloride
RTR	Reliability Test Run
SCE	Safety critical elements
SI	International System Units
SIL	Safety Integrity Level
SRB	Sulphate Reducing Bacteria
SIS	Safety Instrumented System
SWB	Steel Wire Braiding
URV	Upper Range Value
USC	US Customary Units
XLPE	Cross Linked Polyethylene
TEG	Tri-ethylene Glycol
TQ	Top Quartile

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

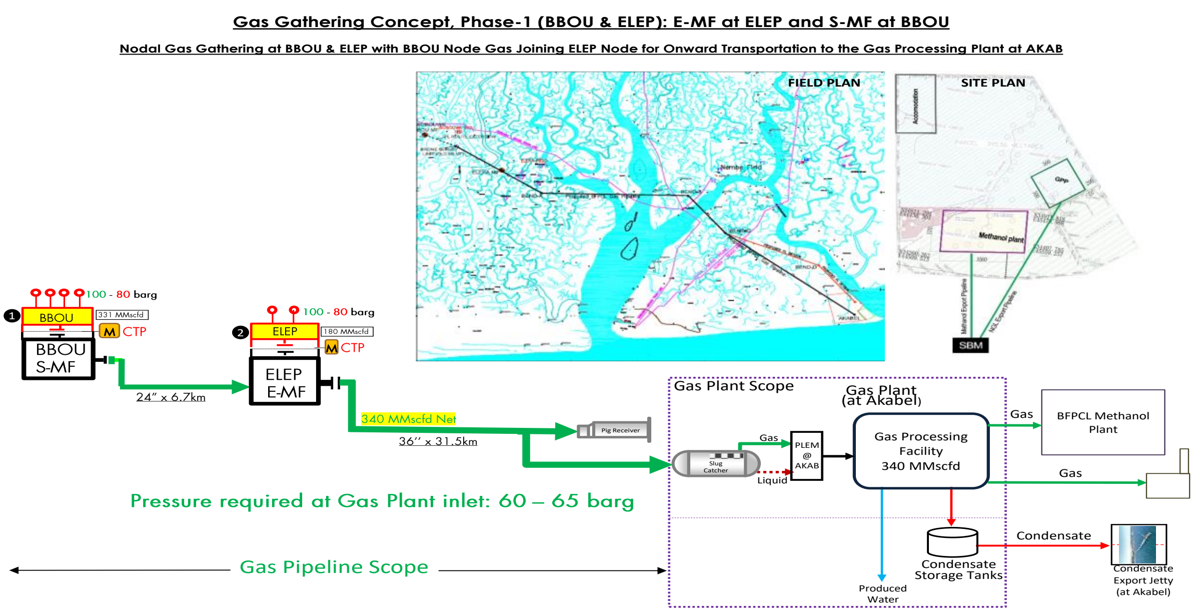

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This purpose of this document is to describe the respective process systems designed for the gathering of raw gas and transportation of the gas from the gas sources at BBOU and ELEP to the gas plant at Akabel.

The FEED shall fully comply with all specified relevant contractual requirements.

Control and Monitoring is described in a separate document titled "Process Control & Safeguarding Philosophy" with Document Number EA-BFPCL-GPMC-GGPL-FEED-PSE-PHY-002.

2. Regulations, Codes And Standards

Document Number	Document Title
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001-3	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/MAT-RPT-001	Pipeline Materials Selection & Corrosion Analysis Study
EA/SPDC-BFPC UGS-GEP/ELE-RPT-001	Manifold Platforms Power Supply Options
EA/SPDC-BFPC UGS-GEP/MEC-RPT-001	Manifold Platforms Concept & Integration with SPDC
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Gas Processing Technology Option
6064/PLRS/001a-PRP	Onshore Pipeline Route Survey for BFCL; Topographical Surveys
6064/PLRS/001PRP	Onshore Pipeline Route Survey for BFCL; Soil Investigation Surveys
40km Gas Supply Pipeline Project ESIA	Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited - Draft Report November 2020.
DEP 31.40.40.38-Gen.	Hydrostatic pressure testing of new pipelines
DEP 33.64.10.ok g10-Gen.	Electrical engineering guidelines
DEP 61.40.20.30-Gen	Welding of pipelines and related facilities

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to.

The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Systems Description

3.1. Overview

The FEED phase 1 Project involves the design of gas gathering, transportation systems and associated systems for delivery of gas from the various fields to the Gas Process Plant at Akabel.

These systems will provide pipeline systems for the transportation of raw gas gathered from the Bubouwe Bou (BBOU) field wells into the BBOU satellite manifold and transported via 24" x 6.7km pipeline to Elepa (ELEP) evacuation manifold where it combines with gas from Elepa field wells. The combined gas is then transported from ELEP to Akabel (AKAB) gas processing plant (GPP) via a 36" x 31.5km pipeline.

The ELEP – AKAP pipeline was initially sized as 28" but latest sizing calculations based on updated fluid characteristics and RoW parameters confirm the size as 36".

3.2. The Battery Limit

The process description will be limited to the following:

3.2.1. Bubouwe Bou (BBOU) Platform

- The terminal flanged ball valve on the manifold collecting well fluids from BBOU Wells.
- The flanged ball valve provided for the tie-in of Opugbene (OPUG) Pipeline on the manifold transporting well fluids to ELEP.

3.2.2. Elepa (ELEP) Platform

- The terminal flanged ball valve on the manifold collecting well fluids from ELEPA Wells.
- The flanged ball valve provided for the tie-in of NUN River Pipeline on the manifold transporting well fluids to AKAB.

3.2.3. Akabel Facility

- The pipeline terminal flange at the inlet to the slug catcher at AKAB.
- The drain outlet flange on the pig receiver at AKAB.

3.3. Description of the Process Systems

This section describes the respective process systems designed for the gathering of raw gas and transportation of the gas from the various gas sources to the gas plant.

3.3.1. Sources of Gas

The transfer of gas custody from the Shell Petroleum Development Company of Nigeria Limited (SPDC) to Brass Fertilizer Company Limited (BFCL) takes place at each field. SPDC is responsible for the gathering of the gas from the field wells through flowlines to each field manifold. After gathering, custody transfer measurement takes place before the gas is delivered through the pipeline to the point of use.

Generally, SPDC will own, operate, and maintain flow and gas quality measuring equipment at Custody Transfer Points on the pipeline network. Gas will be received at Seibou, Nun River, Opugbene, Bubuowe and Elepa fields. Gathered gas at Elepa Manifold Station will be delivered in bulk to the Akabel Gas Processing Plant where fiscal metering will take place

For this Phase of the Project, wet gas from the BBOU and ELEPA Wells are gathered through a header within the SPDC manifold area of the Platforms. The size and description of the Header is not within the scope of this document. The wet gas is transferred to the BFPCL Manifold at the flanged joint Tie-Point Numbered TP 101 on each Platform.

3.3.2. Chemical Injection Skid

The BBOU satellite manifold station has a chemical injection skid installed to inject chemical mainly for corrosion control within the carbon steel pipeline. It functions by dosing the calculated volume of inhibitors (chemicals) per volume of gas (defined). The rate of injection is dependent on the MSDS of the inhibitor. The skid will have capacity for chemical/inhibitor storage for up to 2 – 4 weeks.

The actual chemical to be used for corrosion inhibition, the quantity, pump flowrate and volume of storage tank for chemical injection will be determined during Detailed Engineering Design (DED).

3.3.3. Product Metering

Although, SPDC will own, operate, and maintain flow and gas quality measuring equipment at Custody Transfer Points, a check meter will be installed by BFPCL as the gas enters into the Manifold. This will take place at every CTP on the pipeline network.

The standard approach for wet-gas metering treats the fluid as a gas adopting differential-pressure devices (venturi or orifice) or vortex or ultrasonic meters. The venturi meter has been adopted for this Project because it is more rugged than the orifice in wet-gas service and allows for higher differential pressure operation.

3.3.4. Inlet Pressure Control System

There is no processing, conditioning or phase separation taking place within the Gas Gathering Pipelines and Associated Infrastructure therefore there is no control function (no pressure or level control) within the facilities. There is only monitoring and safeguarding function.

To ensure that the Manifold does not see more than the design pressure of 85-100 barg, a pressure-controlled shutdown system comprising of pressure-controlled shut-down valves (SDV 1001, SDV 1003 and SDV 1004. SDV 1004 applies to Elepa Manifold only) is installed at the inlet of the Manifold.

The shut-down valve will be activated at 100 barg measured by pressure transmitters (PT 1001 & 1002) installed in the Manifold Header.

The shutdown valve is equipped with Gas over oil actuator which uses the wet gas as power, and hydraulic oil as transmission medium to drive the opening and closing of the valve. It's basic control functions include local pneumatic operation, manual hydraulic pump operation, automatic pipe breaking protection, remote electronic control on/off, ESD emergency shutdown, etc.

The pressure class up to downstream of the PCV 1001 is ANSI Class 900 and Class 1500 at BBOU and Elepa Manifolds respectively.

The control schematic diagram for the shut-down valve will be firmed up, and the catalogue provided, during DED.

3.3.5. Manifold Headers

This manifold is an arrangement of pipes and valves for the aggregation and distribution of gas feed from the BBOU and Opugbene field wells. The manifold at BBOU is a 24" ANSI Class 900 header and it is provided with 4 nos. of slot for tie-ins. At Elepa, it is a 36" ANSI Class 900 with 4 nos. of slot for tie-ins. The headers are designed as per ASME B31.3 and equipped with safety instruments in accordance with API 14C requirements.

Safety instruments installed on the Header are Pressure Transmitter PT 1001, PT 1002, PSV 1001 & 1002, TI 1001 and TIT 1001.

The pressure transmitters, temperature transmitters and fire and gas detectors control and activate the emergency shut down valves at the Manifolds.

The pressure safety valve (PSV) is set at 110 barg which is 10% above the maximum operating pressure of 100 barg.

3.3.6. Pig Receiver

There is one Pig Receiver at BBOU (BBOU-A-100) and two at Elepa (ELEP-A-100 and ELEP-A-200). The pig receivers at BBOU-A-100 and ELEP-A-100 are future installations provided for the incoming OPUG-BBOU and NUNR-ELEP bulk line pigging activities respectively. The pipelines are designed to perform intelligent pigging operation, designed in accordance to ASME B31.8 and rated ANSI Class 900 with saddle supports. They are equipped with valves, closure door and connections to closed drain system. The Pig receiver is also equipped with pig detector (ZI 1001 and ZI 1002) and pressure indicators (PI-1002 & 1003).

The size and dimensions of the Pig Receivers are as follows:

Equipment	Major Barrel Size	Minor Barrel Size	Major Barrel	Minor Barrel

			Length	Length
BBOU-A-100	28 inch	24 Inch	5700 mm	5700 mm
ELEP-A-100	40 inch	36 Inch	6600 mm	5300 mm
ELEP-A-200	28 inch	24 Inch	5700 mm	5700 mm

3.3.7. Pig Launcher (BBOU-A-200)

There is one Pig Receiver at both BBOU (BBOU-A-200) and Elepa (ELEP-A-300). The outgoing pig receiver (BBOU-A-200) is installed for the pigging of the BBOU-ELEP pipeline while the ELEP-A-300 is installed for the ELEP-AKABEL Pipeline.

The Launchers are designed in accordance to ASME B31.8 and rated ANSI Class 900 with saddle supports. They are equipped with valves, closure door and connections to closed drain system. The Pig Launchers are also equipped with pig detector (ZI 1003 and ZI 1004) and pressure indicators (PI-1007, PI 1008 and PI 1009).

The size and dimensions of the Pig Launchers are as follows:

Equipment	Major Barrel Size	Minor Barrel Size	Major Barrel Length	Minor Barrel Length
BBOU-A-200	28 inch	24 Inch	5700 mm	1500 mm
ELEP-A-300	40 inch	36 Inch	6600 mm	1500 mm

3.3.7.1. Pipeline

The pipelines transport well fluids from BBOU to Elepa and from Elepa to Akabel. The BBOU-ELEP 6.7km pipeline transport 331 MMSCFD of wet gas through a 24" Class 900 Pipeline. On the other hand, the ELEP-AKAB pipeline is a 36" x 31.5km, Class 900 pipeline transporting raw gas at a flowing rate of 386 MMSCFD. Both Pipelines have an operating pressure of 100 barg and made of API 5L – X70 material. They are equipped with safety instruments in compliance with API 14C requirements. Both pipelines are piggable.

3.3.7.2. Closed Drain Vessel

The closed drain vessel (BBOU-CDT-001 and ELEP-CDT-001) mainly collects the sludge during pigging operation and also serves as knock out drum for the vent system. The level in the vessel is monitored in the control room, while a local level gauge is provided for monitoring the level locally. The high-level indication in the control room or field will call for evacuation of the vessel. The evacuation of the vessel will be planned and done through evacuation barges that connect to the vessel through a hose.

The tank has a diameter of 1,000mm and shell length of 3,000mm (T/T) and a design pressure of 7barg. It is also equipped with pressure and temperature transmitters.

3.3.7.3. Open Drain Tank

The open drain tank (BBOU-ODT-001 and ELEP-ODT-001) receives collected from the drip pans of pig trap skids. It is an atmospheric tank with baffles to allow preliminary separation of oil, water, and solid particles. The tank has the dimension of 2,000mm x 1,000mm x 1,000mm for its length, width and diameter respectively. Its design pressure and temperature are both atmospheric.

3.3.7.4. Vent System

3.3.7.4.1. Vent Header

The vent header is a 6" line that collects fluids from BDV 1001, PSV 1001, PSV 1002, PSV 1003, and PSV 1004. It is rated for ANSI Class 150. Its installation is such that it drains, without pockets, to the closed drain vessel (BBOU-CDT-001 and ELEP-CDT-001).

3.3.7.4.2. Elevated Vent Stack

The elevated vent stack installed to handle vent gas generated during process upsets, planned routine shutdowns and during emergency shutdowns that could lead to manual depressurization and blowdown operations. The vent stack is connected to the closed drain vessel (BBOU-CDT-001 and ELEP-CDT-001) and located at a safe location within the manifold stations.